When Careers Stall: Early-Career Job Insecurity and Fertility over the Life Cycle

Job Market Paper

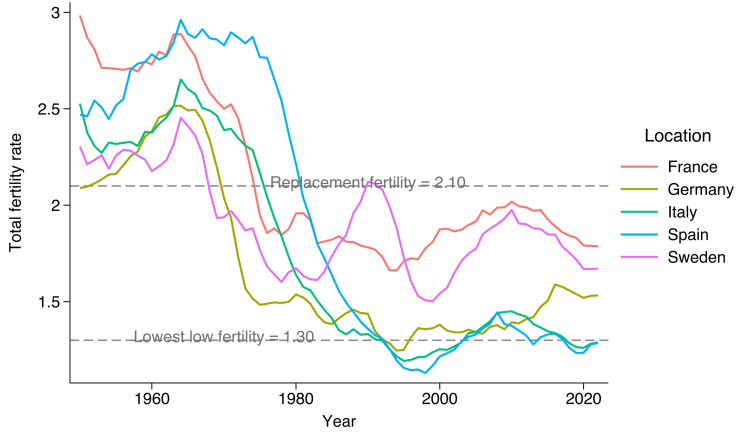
Mario Martínez-Jiménez Judit Vall Castelló

Imperial College London Stanford University Universitat de Barcelona

Applied Economics Conference
Belgrade, Serbia
2026

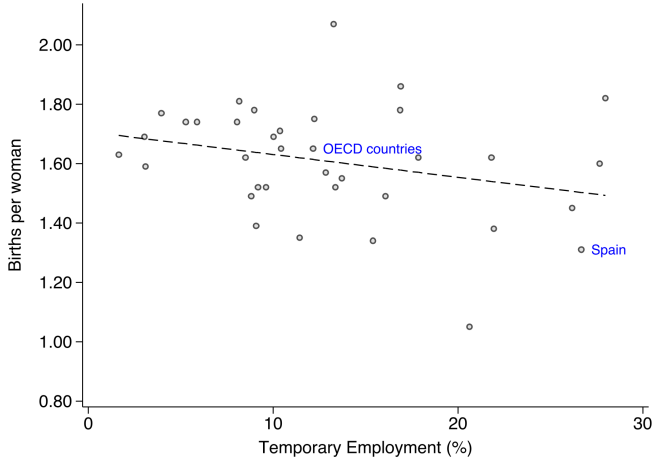
Motivation: fertility has fallen across Europe

European fertility has remained below replacement for decades; Spain and Italy reached the "lowest-low" range.



Source: United Nations World Population Prospects; JMP Appendix A.

Motivation: temporary employment and fertility move together



Across OECD countries, greater reliance on temporary employment is associated with lower fertility. This descriptive relationship motivates the causal design; it is not itself causal evidence.

Source: OECD 2017; JMP Figure 1.

This Paper

Question. Can greater exposure to fixed-term contracts at labor-market entry have lasting effects on fertility?

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Economic margins

- **Labor outcomes:** recorded employment entry (**extensive**), days worked (**intensive**), non-permanent employment, and contribution-based income.
- **Fertility outcomes:** births per woman, combining motherhood (**extensive**) and additional births (**intensive**); maternal age at all and first births (**timing**).

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Our approach

- **Quasi-experiment:** nearby cohorts reached expected entry age before or after Spain's 1984 reform, changing exposure to job insecurity at the start of a career.
- **Datasets:** Birth records, MCVL employment histories, and Fertility Surveys.
- **Reduced form:** trace entry exposure into subsequent fertility; assess dynamics and selection.
- **Model:** women weigh a birth now against future job prospects and declining birth success.

Related Literature and Contributions

1. **Labor-Market Institutions and Career Scarring**
   Labor-market institutions and entry conditions shape subsequent employment and earnings: Blanchard and Landier (2002); García-Pérez et al. (2019); Oreopoulos et al. (2012).

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2. **Employment Risk and Fertility**
   Job loss, temporary contracts, and perceived insecurity shape fertility decisions and intentions: Adserà (2004); Del Bono et al. (2012); Auer and Danzer (2016); Vignoli et al. (2020).

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3. **Dynamic Fertility Choice and Preference Heterogeneity**
   Quantitative models connect fertility to earnings risk, career costs, and household preferences: Sommer (2016); Adda et al. (2017); Doepke and Kindermann (2019); Guner et al. (2024).

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3. **Dynamic Fertility Choice and Preference Heterogeneity**
   ⇒ **This paper:** embeds desired family size, religiosity, and gender-role attitudes in a dynamic fertility model.

Outline

1. **Motivation and Contribution**
2. Setting and Design
3. Reduced-Form Results
4. Structural Model
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Setting: Spanish Labor Market and the 1984 Reform

Before 1984
Regulated labor market
Permanent jobs dominate
Strong firing protection
Temporary work restricted

→

1984 reform
Fixed-term contracts
Allowed for all activities
Lower dismissal costs
Shorter contract duration

→

After 1984
Dual labor market
Permanent: stable, protected
Temporary: flexible, cheaper, repeated short spells

Channels and outcomes for exposed women

Most exposed
young,
low-skilled

Temporary share (1998)
Women: **34.4%**
Age 15–24: **73.0%**
Low education: **37.7%**

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Temporary jobs: 70%
Days worked: −4.9%
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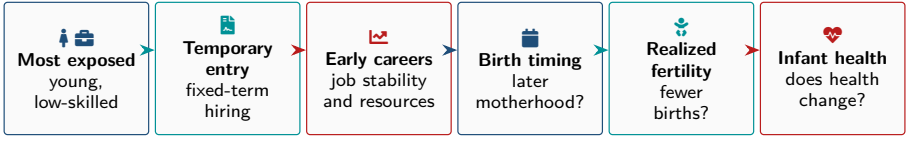
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Setting: temporary employment rises sharply after the reform



Temporary employment rose from about **2.2% in 1983** to **20% in 1987**, becoming a central feature of Spain's dual labor market.

Dashed vertical line marks the 1984 reform.

Data and Exposure Design

Source	Coverage and role
Birth records	All registered births in Spain, 1980–2017, all maternal nationalities. Birth quantity, timing, first deliveries and perinatal outcomes.
Fertility Surveys	Official respondent weights. Pooled 1999/2018 child counts include childless women; 2018 mechanism evidence.
MCVL 2006	Recorded affiliation, career stability, employee days and contribution-based income. Separate entry-age and attached-career samples.
WVS / ISSP	Descriptive cultural context; model preference gradients come from the 2018 Fertility Survey.

Low-skill cohorts: 1965Q1–1972Q4, cutoff 1968Q4 (expected entry at 16). High-skill cohorts: 1962Q1–1969Q4, cutoff 1966Q1 (expected entry at 18). Skill proxies are source-specific.

Exposure is assigned at expected labor-market entry; fertility is followed afterward. Individual contract choices do not define treatment.

Fertility Outcomes: What Do We Measure?

Quantity: Birth records

$$\underbrace{Y_{cps}}_{\text{births per woman, cohort-level}} = \frac{\overbrace{B_{cps}}^{\text{registered infants, 1980–2017}}}{\underbrace{\widehat{D}_{cps}}_{\text{estimated female cohort size}}}.$$

A cell is a mother's birth-cohort quarter $c \times$ province $p \times$ skill s. The denominator includes women who never give birth.

How the denominator is built →

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Timing and health: Birth records

Maternal age, first-delivery age, and infant health are observed among realized births. First deliveries retain every infant from a multiple birth.

The cohort ratio and individual child counts capture related quantity margins; they are not the same measure of completed fertility.

How the denominator is built →

Descriptive Statistics: Fertility across Datasets

Weighted means in the analysis samples

Outcome	Low skill	High skill
Birth records, 1980–2017		
Births per woman (cohort-level)	1.443	1.112
Maternal age at birth (years)	28.068	31.576
Maternal age at first delivery (years)	26.031	30.232
Fertility Surveys, 1999/2018: including childless women		
Biological children per respondent	1.319	1.158
Childless (%)	25.9	33.3
Maternal age at birth (years)	26.185	29.764
Maternal age at first birth (years)	24.428	28.274
Fertility Survey, 2018: mechanism outcomes		
Desired children exceed biological children (%)	33.8	41.5
Temporary contract among current employees (%)	28.0	14.9

Notes: Birth records use cohort-size weights for quantity and equal infant weights for ages; surveys use respondent weights. Birth-age means condition on observed births. Samples and skill proxies differ; the 1999 wave need not capture completed fertility.

Perinatal descriptive statistics →

First Stage: Early Recorded Affiliation

$$Y_{ic}^{a} = \alpha + \beta\,\text{After}_c + \gamma\,\text{Trend}_c + \delta\,(\text{After}_c \times \text{Trend}_c) + \pi_q + \varepsilon_{ic}.$$

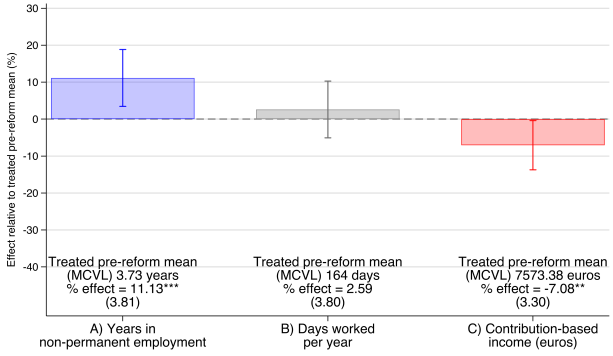
$$Y_{ic}^{a} = \mathbf{1}\{\text{first affiliation by age } a\}, \qquad a \in \{18, 19\}.$$

	(1)	(2)	(3)	(4)
	By age 18		By age 19	
	Baseline	Transition	Baseline	Transition
After	0.035	0.035	0.024	0.050*
	(0.033)	(0.042)	(0.027)	(0.029)
Trend	−0.012***	−0.012***	−0.004	−0.006***
	(0.002)	(0.003)	(0.002)	(0.002)
After × Trend	0.023***	0.025***	0.012***	0.014***
	(0.003)	(0.004)	(0.003)	(0.003)
Mean dep. var.	0.611	0.611	0.816	0.816
Pre-reform mean	0.590	0.590	0.782	0.782
R-squared	0.020	0.020	0.015	0.016
Observations	4,474	4,474	4,474	4,474
Birth-quarter controls	Yes	Yes	Yes	Yes
Transition-cohort indicators	No	Yes	No	Yes

Note: Estimated separately by age threshold. π_q: birth-quarter effects. Transition specifications add indicators for cohorts 1968Q3 and 1968Q4; standard errors cluster by cohort quarter.

First Stage: Contractual Instability and Contribution-Based Income

$$Y_i = \alpha + \beta(\text{Post}_c \times \text{LowSkill}_i) + \theta\text{LowSkill}_i + \delta\text{Post}_c + g_s(\text{Trend}_c, \text{Post}_c) + \kappa U_i + \lambda_r + \pi_q + \varepsilon_i$$



Note: Y_i is alternatively years in non-permanent employment, days worked per potential year, or contribution-based income. The figure reports β relative to the corresponding treated pre-reform mean. g_S allows skill-specific linear cohort slopes before and after the cutoff. U_i is regional unemployment at first retained affiliation; λ_r and π_q denote entry-region and birth-quarter effects.

Non-permanent years: $+0.415$ (SE 0.142); days per potential year: $+4.260$ (6.244); contribution-based income: EUR -535.821 (249.859). No detectable change in days worked.

Empirical Strategy: within-cohort difference in differences

$$y_{ucpsw} = \alpha + \beta_1 r_{cs} + \beta_2 (r_{cs} \times Post_{cs}) + \beta_3 Post_{cs} + \beta_4 LowSkill_s + \boxed{\beta_5 (LowSkill_s \times Post_{cs})} + \eta_p + \lambda_w + \varepsilon_{ucpsw}.$$

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- Birth records: province effects, no survey-wave term; birth-count weights for quantity and equal infant weights for age/health. Surveys: region and wave effects, respondent weights.
- Standard errors cluster by mother's birth-cohort quarter; the main Birth-record sample has 44 clusters.

Identification and Diagnostics

Diagnostics: annual event studies, synthetic comparisons, alternative cohort trends and windows, and survey-calibrated selection sensitivities.

- **Dynamic patterns:** annual event studies show when the changes emerge relative to the last pre-reform cohort bin.
- **Synthetic comparison:** match pre-reform fertility using high-skilled provincial paths, then keep the comparison fixed. Fewer births also appear against this benchmark.
- **Sensitivity checks:** alternative trends and cohort windows retain fewer births and later childbearing; magnitudes and precision vary. [Specifications]
- **Across datasets and measures:** fewer children also appear in the Fertility Surveys (including childless women) and in raw/log birth counts. [Survey comparison] [Alternative outcomes]

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Fertility Quantity: Fewer Realized Births

Annual event-study specification

$$Y_{cps} = \alpha + \eta_p + \theta L_s + \sum_{m \neq -1} \gamma_m \, \mathbf{1}\{a_{cs} = m\} + \sum_{m \in \mathcal{M}} \beta_m \, \mathbf{1}\{a_{cs} = m\} L_s + \varepsilon_{cps}.$$

a_{cs}: annual bins of relative cohort position; $m = -1$ is the reference. $L_s = 1$ for low skill. γ_m: common bin effects; β_m: low-skill contrasts, $\mathcal{M} = \{-4, -3, -2, 0, 1, 2, 3\}$.

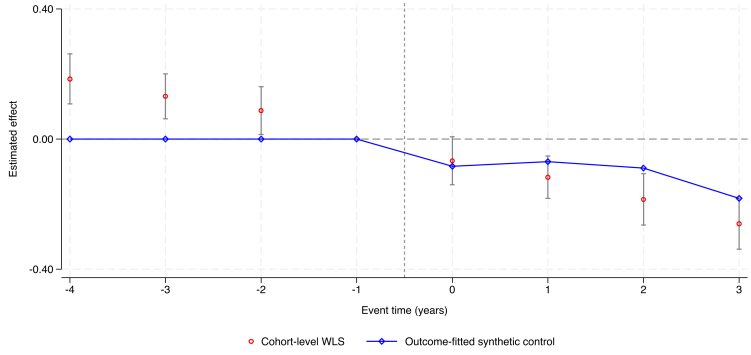
	DD	Annual event study
Estimated coefficient	-0.257^{***}	-0.157^{***}
	(0.018)	(0.031)
Effect (% of treated pre mean)	-15.2	-9.3
Observations	3,200	3,200
R-squared	0.456	0.459

Notes: Cohort-level birth intensity, not unique-woman completed fertility. DD uses the preceding specification. Event-study estimate: average of $\beta_0, \ldots, \beta_3$. Birth-count weights; cohort-quarter clustered standard errors in parentheses; *** $p < .01$.

Denominator construction →

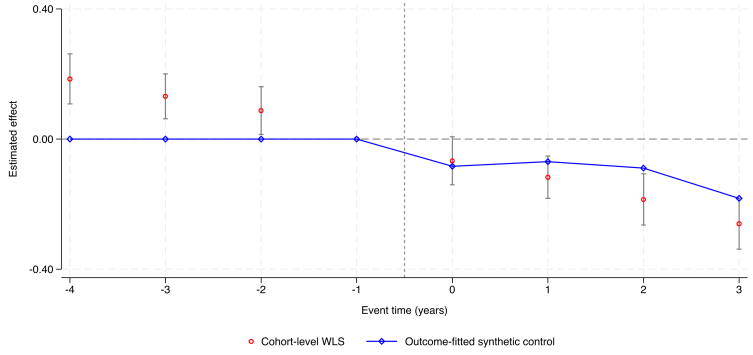
Survey quantity check →

Fertility Quantity: Annual and Synthetic Comparisons



Fertility Quantity: Annual and Synthetic Comparisons

Synthetic control. Combine high-skilled provincial fertility paths to match low-skilled fertility before the reform. Keep those weights fixed and compare the paths afterward.



Why it builds confidence. With the comparison fixed in advance, the post-reform data again show fewer births. The decline appears across DD (-15.2%), annual estimates (-9.3%), and the synthetic gap (-6.3%).

Fertility Timing: Later Motherhood

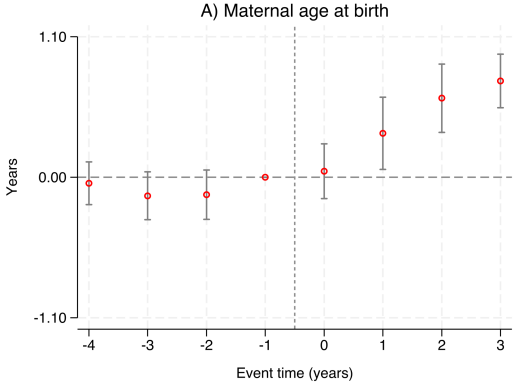
	Maternal age at birth	First-delivery age
Panel A: DD estimates		
Post × low-skilled mother	0.526***	0.441***
	(0.067)	(0.060)
Treated pre-reform mean	27.428	25.368
Effect (% of treated pre-reform mean)	1.9	1.7
Observations	3,382,191	1,880,215
R-squared	0.107	0.154
Panel B: Selection sensitivity		
Conditional: lower bound	0.055	0.036
Conditional: upper bound	0.982	0.877
Joint-normal: lower bound	0.165	0.109
Joint-normal: upper bound	0.885	0.773

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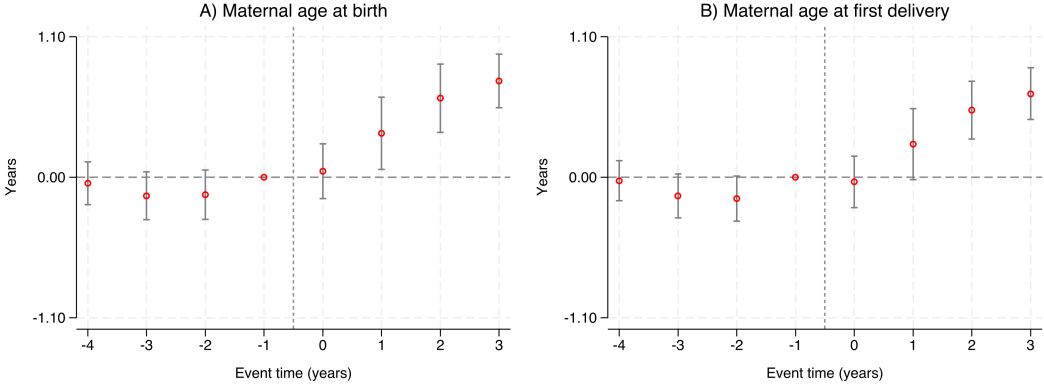
The **Lee-style** exercise allows the reform to reduce selection into motherhood. The **conditional Lee-style** exercise assumes weakly decreasing selection into motherhood and trims the comparison group within cohort-quarter–province cells; the **joint-normal exercise** adds a joint normality assumption for outcomes and selection.

Annual Event Studies: Maternal and First-Delivery Age



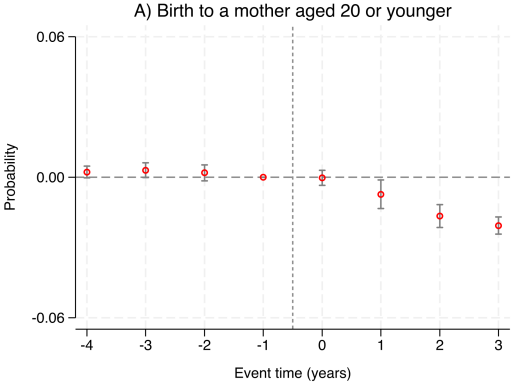
Maternal age: $+0.526$ years.

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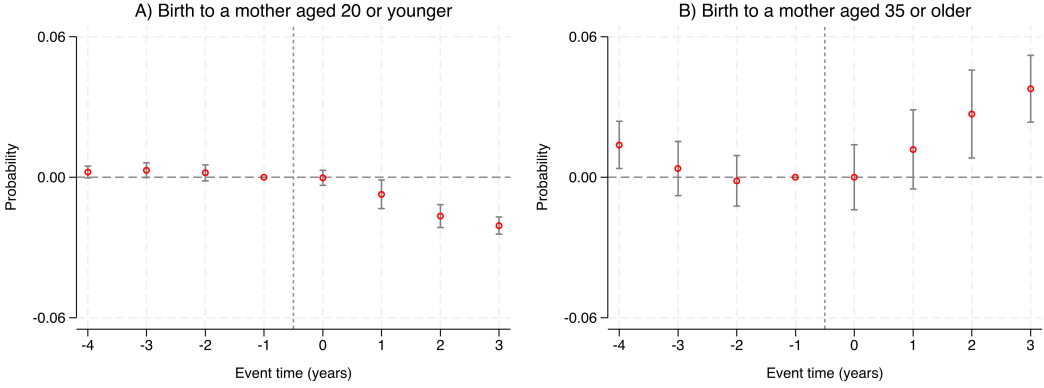
Maternal age: $+0.526$ years. First-delivery age: $+0.441$ years.

Maternal-Age Thresholds Among Realized Births



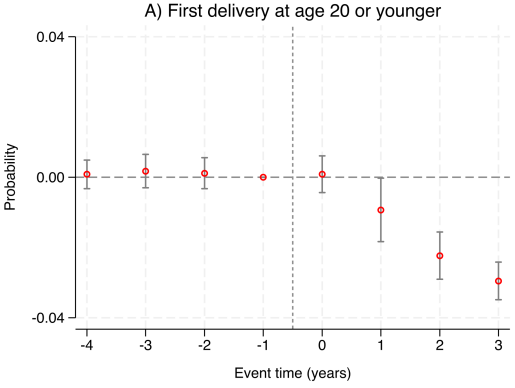
DD: age ≤ 20, -1.3 pp.

Maternal-Age Thresholds Among Realized Births



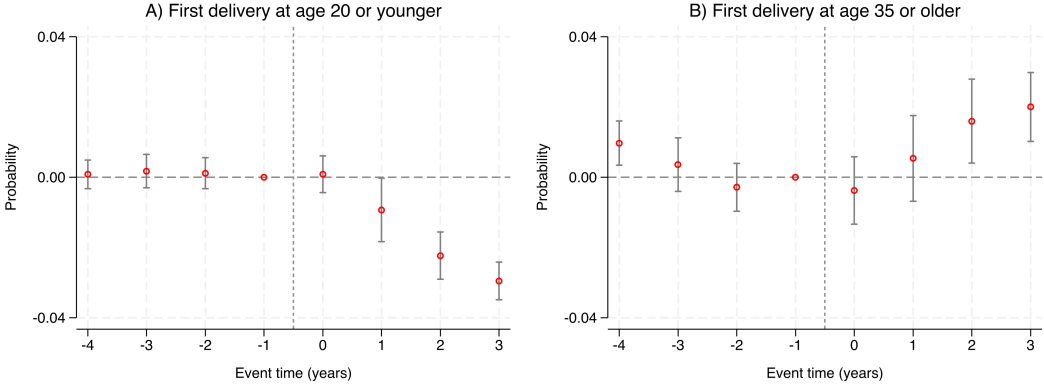
DD: age ≤ 20, -1.3 pp. Age ≥ 35, $+1.5$ pp; age ≥ 40, $+0.8$ pp.

Age Thresholds at First Delivery



DD: age ≤ 20, -1.6 pp.

Age Thresholds at First Delivery



DD: age ≤ 20, -1.6 pp. Age ≥ 35, $+0.7$ pp; age ≥ 40, $+0.5$ pp.

Perinatal Outcomes: No General Deterioration

	Birthweight	Low BW	High BW	Gest. age	Preterm	C-section
Panel A: First deliveries: DD level effects						
Exposed cohort × low-skilled	3.788**	-0.001	0.001**	0.066***	-0.012***	0.044***
	(1.585)	(0.001)	(0.000)	(0.008)	(0.001)	(0.009)
Treated pre mean	3196.912	0.071	0.038	38.931	0.118	0.451
Mean dep. var.	3189.830	0.075	0.038	39.020	0.098	0.408
Effect (% of treated pre mean)	0.1	-1.8	3.1	0.2	-10.3	9.7
Observations (millions)	1.71	1.71	1.71	1.48	1.48	0.08
R-squared	0.004	0.001	0.001	0.007	0.017	0.023
Panel B: Selection sensitivity						
Conditional: lower bound	-49.670	-0.005	-0.001	-0.182	-0.016	0.023
Conditional: upper bound	46.095	0.036	0.022	0.184	0.026	0.069
Joint-normal: lower bound	-32.754	-0.005	-0.001	-0.079	-0.016	0.021
Joint-normal: upper bound	40.338	0.045	0.037	0.211	0.034	0.071

Mechanisms: Employment Insecurity and Unmet Fertility

Outcome	DD (raw)	SE	Pre mean	BH p-value
Temporary contract, observed employees	0.157^{**}	0.071	0.247	0.1656
Unmet desired fertility	0.279^{**}	0.106	0.294	0.0579
Married	0.018	0.101	0.744	0.8610
Home ownership	-0.094	0.070	0.792	0.4739
No childcare used, child under 14	-0.205	0.231	0.396	0.6352

Mechanisms: Employment Insecurity and Unmet Fertility

Multiple testing: Benjamini–Hochberg (BH) adjustment within each five-outcome economic or preference/support family.

Outcome	DD (raw)	SE	Pre mean	BH p-value
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No childcare used, child under 14	-0.205	0.231	0.396	0.6352

Only unmet desired fertility remains significant at the 10% level after BH adjustment. Exposed women are more likely to have fewer biological children than they desire.

Household Evidence: Recorded Father Skill

	(a) Birth quantity (cohort-level)	(b) Maternal age at birth	(c) First-delivery age
Panel A. Mother-side benchmark			
Post × low-skilled mother	−0.257***	0.526***	0.441***
	(0.018)	(0.067)	(0.060)
Panel B. Father-side skill among observed births			
Post × low-skilled mother × low-skilled father	−0.182***	0.406***	0.331***
	(0.013)	(0.064)	(0.058)
Post × low-skilled mother × high-skilled father	−0.047***	0.995***	0.938***
	(0.014)	(0.085)	(0.084)
Treated pre mean: low-skilled father	1.197	27.023	25.021
Observations	4,800	3,148,416	1,737,176
R-squared	0.736	0.121	0.167

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Outline: Structural Model

4. Structural Model
- Specification
- Estimation
- Model Fit
- Counterfactuals

Structural Model: What It Adds

Motivation: The reduced form identifies the fertility response, but cannot show why temporary work changes lifetime fertility.

Intuition: A woman compares a birth now with waiting for a more secure job, while fecundity and the remaining horizon decline with age.

- **Organize dynamics:** labor states, resources, parity, spacing, and cumulative temporary exposure evolve *over the life cycle*.
- **Separate channels:** entry composition, persistent transitions, earnings, and a direct insecurity cost.
- **Counterfactual experiments:** decompose the 1984 reform and test marginal reductions in temporary exposure.

Key: The model combines observed labor dynamics with a forward-looking fertility choice.

Model questions and evidence →

Specification: One Annual Fertility Decision

Each year: observe her situation, try for a child or wait, then move to next year.

repeat at age $a + 1$

1. Her situation

$x_a = (s, e_a, n_a, g_a, h_a)$: skill, job status, number of children, time since the last birth, and past temporary-work years (h_a, capped at 2).

States and transitions →

Specification: One Annual Fertility Decision

Each year: observe her situation, try for a child or wait, then move to next year.

repeat at age $a + 1$



2. Her decision

Taking her circumstances into account, try for a child, $d_a = 1$, or wait, $d_a = 0$.

1. Her situation

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repeat at age $a + 1$

3. Birth outcome

An attempt results in a live birth with probability $q(a)$, which varies with age.

2. Her decision

Taking her circumstances into account, try for a child, $d_a = 1$, or wait, $d_a = 0$.

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$x_a = (s, e_a, n_a, g_a, h_a)$: skill, job status, number of children, time since the last birth, and past temporary-work years (h_a, capped at 2).

Age and birth success (G2) →

States and transitions →

Specification: One Annual Fertility Decision

Each year: observe her situation, try for a child or wait, then move to next year.

repeat at age $a + 1$

4. Next year

Job status may change, following $P_{s,a}$. A birth adds one child and restarts the spacing clock. Temporary work adds to h_a, up to 2; a permanent job does not erase past exposure.

3. Birth outcome

An attempt results in a live birth with probability $q(a)$, which varies with age.

2. Her decision

Taking her circumstances into account, try for a child, $d_a = 1$, or wait, $d_a = 0$.

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$x_a = (s, e_a, n_a, g_a, h_a)$: skill, job status, number of children, time since the last birth, and past temporary-work years (h_a, capped at 2).

Age and birth success (G2) →

States and transitions →

Bellman Equation: Try Now or Wait?

Compare trying now with waiting: what matters today and what may happen next?

$$V_a(x) = E_\varepsilon \left[\max_{d \in \{0,1\}} \left\{ \underbrace{u_a^d(x)}_{\text{Today}} + \underbrace{\varepsilon_{ad}}_{\text{Choice shock}} + \underbrace{\beta E[V_{a+1}(x') \mid x, d]}_{\text{Future}} \right\} \right]$$

Choice: d

$d = 0$: wait this year. $d = 1$: attempt a birth this year.

She considers her job, family, and personal reasons to try or wait.

Neither a birth nor a better job next year is guaranteed.

Exact recursive problem →

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Today: resources plus the expected benefit of another child

$$u_a^d(x) = \log c_a^0(x) + d\, q(a)\, \Delta u_a(x).$$

$c_a^0(x)$: resources left after supporting existing children.

Trying adds the chance $q(a)$ of receiving the birth payoff $\Delta u_a(x)$.

Waiting or an unsuccessful attempt brings no new-birth payoff.

Age and birth success (G2) →

Exact recursive problem →

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Today: resources plus the expected benefit of another child

$$u_a^d(x) = \log c_a^0(x) + d\, q(a)\, \Delta u_a(x).$$

Another child changes today's payoff in four ways.

$$\Delta u_a(x) = \text{child value} + \text{preferences} - \text{costs} + \text{resource change}.$$

These four parts apply when a child is born.

She chooses whether to try, d; success occurs with probability $q(a)$.

Age and birth success (G2) →

Exact recursive problem →

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$$\Delta u_a(x) = \textbf{child value} + \text{preferences} - \text{costs} + \text{resource change}.$$

1. Child value

$$\alpha_0 + \alpha_2 \mathbf{1}\{n = 1\} + \alpha_3 \mathbf{1}\{n = 2\}$$

α_0: baseline child value, including the value of the first child.

α_2: extra value for the second child; total $\alpha_0 + \alpha_2$.

α_3: extra value for the third child; total $\alpha_0 + \alpha_3$.

Full birth payoff →

Bellman Equation: Try Now or Wait?

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2. Observed preferences

$$\alpha_D D + \alpha_R R + \alpha_G G$$

D: desired family size; α_D: how strongly it relates to child value.

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R: religiosity; α_R: how strongly it relates to child value.

G: traditional gender-role attitudes; α_G: their link to child value.

Preferences stay fixed. These relationships are not causal effects of beliefs.

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3. Timing, career, and insecurity costs

$$c_A(a) + c_H \mathbf{1}\{s = H\} \omega_H(a) + c_T h$$

$c_A(a)$: costs of having children early (c_E) or late (c_L).

Full birth payoff →

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$c_A(a)$: costs of having children early (c_E) or late (c_L).

c_H: strength of high-skill career costs; $\omega_H(a)$: how these vary with age, based on pre-reform resources.

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c_H: strength of high-skill career costs; $\omega_H(a)$: how these vary with age, based on pre-reform resources.

$c_T h$: cost of past temporary work, beyond current resources and job status. c_T is the cost per year; h counts up to two years.

Full birth payoff →

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$$\Delta u_a(x) = \text{child value} + \text{preferences} - \text{costs} + \textbf{resource change}.$$

4. Resource change

$$\log c_a^1(x) - \log c_a^0(x)$$

c_a^0: resources without another child.

c_a^1: resources after supporting one more child.

Another child leaves fewer resources for other needs.

Full birth payoff →

Bellman Equation: Try Now or Wait?

Compare trying now with waiting: what matters today and what may happen next?

$$V_a(x) = E_\varepsilon \left[\max_{d \in \{0,1\}} \left\{ \underbrace{u_a^d(x)}_{\text{Today}} + \underbrace{\varepsilon_{ad}}_{\text{Choice shock}} + \underbrace{\beta E[V_{a+1}(x') \mid x, d]}_{\text{Future}} \right\} \right]$$

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4. Resource change

$$\log c_a^1(x) - \log c_a^0(x)$$

$c_a^0 = \max\{\underline{c}, y_{s,a,e} - \kappa n\}, \quad c_a^1 = \max\{\underline{c}, y_{s,a,e} - \kappa(n+1)\}.$

$y_{s,a,e}$: resources approximated using contribution records.

κ: annual resource cost per child; $\underline{c}$: minimum resource level.

Child costs and the minimum resource level are set before fitting fertility choices.

Full birth payoff →

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Future: the value of next year's opportunities

$$\beta E[V_{a+1}(x') \mid x, d]$$

$\beta = 0.96$: the weight placed on next year's opportunities.

Waiting may bring a better job, but leaves less time to have children.

The model follows annual fertility decisions from ages 16 to 40.

Age and birth success (G2) →

Exact recursive problem →

Bellman Equation: Try Now or Wait?

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Future: the value of next year's opportunities

$$\beta E[V_{a+1}(x') \mid x, d]$$

If she tries, allow for either outcome: a birth or no birth.

$$\beta\Big[q(a)E\{V_{a+1}(x'_B)\} + (1 - q(a))E\{V_{a+1}(x'_0)\}\Big].$$

x'_B: her situation after a birth; x'_0: her situation without a birth.

Both outcomes allow job status to change, using $P_{s,a}$, and add any new temporary work to h.

If she waits, there is no birth this year.

Age and birth success (G2) →

Exact recursive problem →

Bellman Equation: Try Now or Wait?

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Choice shock: personal reasons to try or wait

ε_{a0} : reasons to wait, ε_{a1} : reasons to try.

She knows these personal reasons; we do not observe them.

Women with the same observed circumstances can therefore choose differently.

Exact recursive problem →

Bellman Equation: Try Now or Wait?

Compare trying now with waiting: what matters today and what may happen next?

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Choice shock: personal reasons to try or wait

$$\Pr(d = 1 \mid x) = \Lambda\left(v_a^1(x) - v_a^0(x)\right), \qquad \Lambda(z) = \frac{e^z}{1 + e^z}.$$

$v_a^d = u_a^d + \beta E[V_{a+1} \mid x, d]$: the value of trying or waiting before personal reasons enter.

The more attractive trying is relative to waiting, the more likely she is to try.

Birth probability = chance of trying × chance of success: $q(a) \Pr(d = 1 \mid x)$.

Age and birth success (G2) →

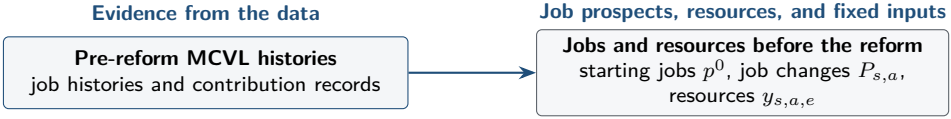
Exact recursive problem →

Outline: Structural Model

4. Structural Model
- Specification
- Estimation
- Model Fit
- Counterfactuals

What Are We Estimating?

1. Estimate job prospects and resources; set the remaining inputs

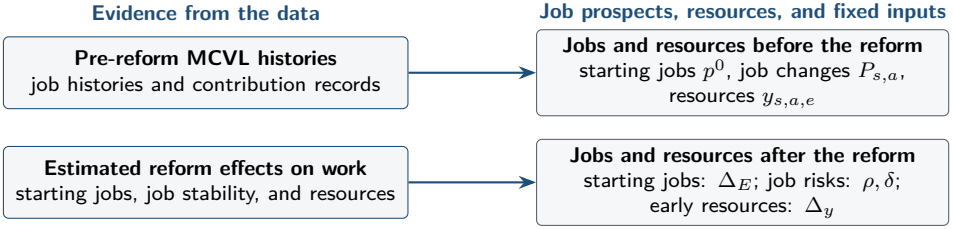


First use employment records to estimate job prospects and resources.

Inputs and transition estimates →

What Are We Estimating?

1. Estimate job prospects and resources; set the remaining inputs



Δ_E changes low-skilled women's starting jobs at age 16; high-skilled entry remains at 18.

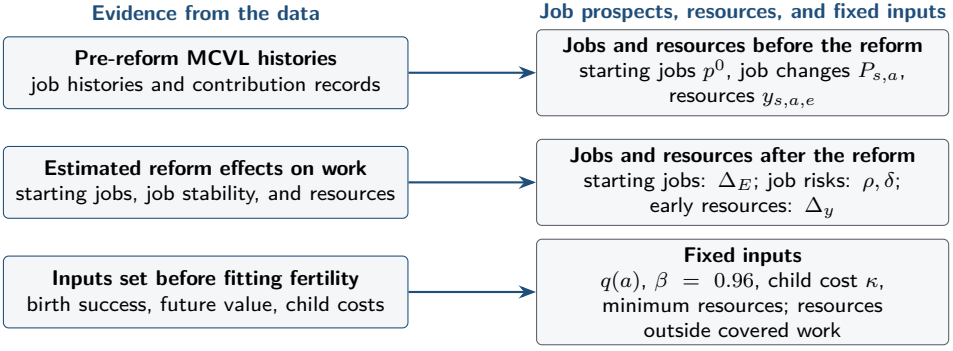
Fit ρ, δ to job risks and Δ_y to early resources before fitting fertility.

The appendix explains how each employment sample maps into the model.

Inputs and transition estimates →

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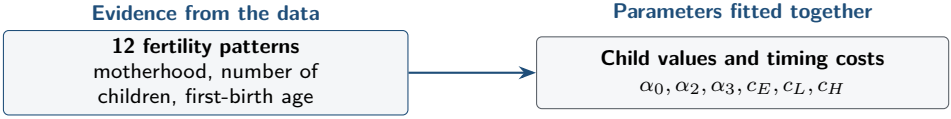
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Birth-success calibration (G2) →

Inputs and transition estimates →

What Are We Estimating?

2. Hold those inputs fixed; estimate fertility preferences and costs



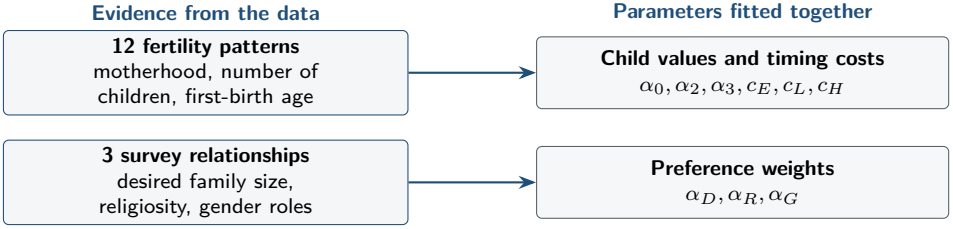
Arrows show which data are most informative. All ten parameters are fitted together.

Birth-success calibration (G2) →

Estimator and identification →

What Are We Estimating?

2. Hold those inputs fixed; estimate fertility preferences and costs



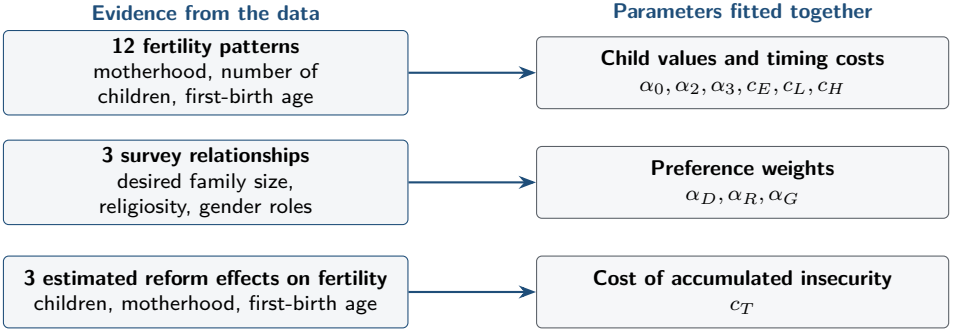
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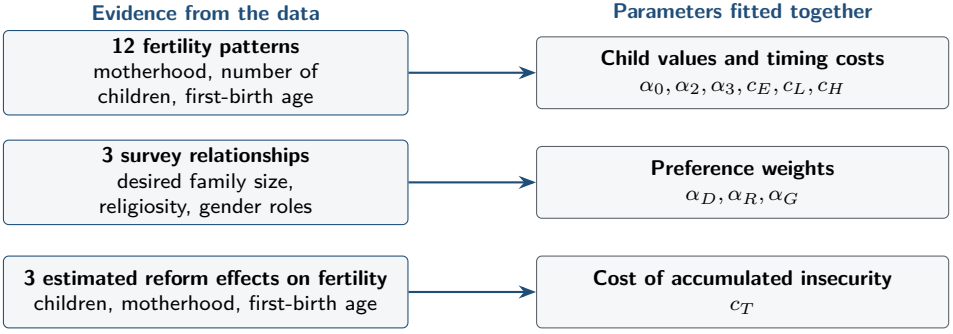
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Estimator and identification →

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2. Hold those inputs fixed; estimate fertility preferences and costs



Match 18 features of the data using 10 fertility parameters.

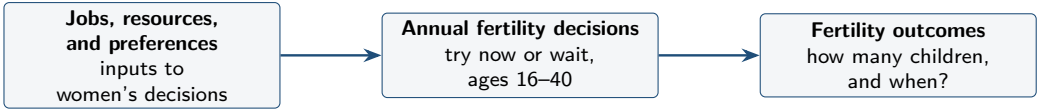
These parameters govern whether women try for a child and how their families grow.

Birth-success calibration (G2) →

Estimator and identification →

What Do We Get from the Model?

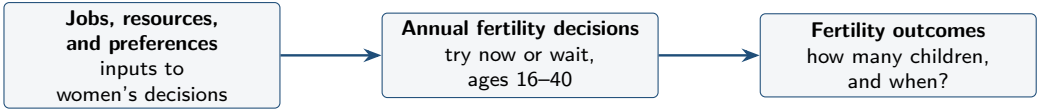
The model follows women's work and fertility choices over time. We check whether it reproduces the data, then change individual channels to understand their roles.



Model fit details →

What Do We Get from the Model?

The model follows women's work and fertility choices over time. We check whether it reproduces the data, then change individual channels to understand their roles.



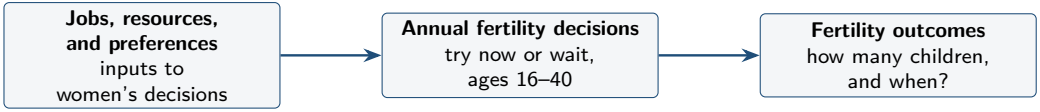
These predictions help answer three questions:

Model fit
Do predictions match the data?

Model fit details →

What Do We Get from the Model?

The model follows women's work and fertility choices over time. We check whether it reproduces the data, then change individual channels to understand their roles.



These predictions help answer three questions:

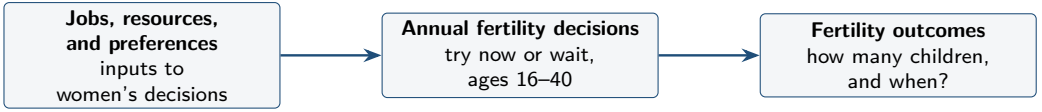
Model fit
Do predictions match the data?

Mechanisms
Which channels explain the change?

Model fit details →

What Do We Get from the Model?

The model follows women's work and fertility choices over time. We check whether it reproduces the data, then change individual channels to understand their roles.



These predictions help answer three questions:

Model fit	**Mechanisms**	**Lower insecurity**
Do predictions match the data?	Which channels explain the change?	How much fertility could recover?

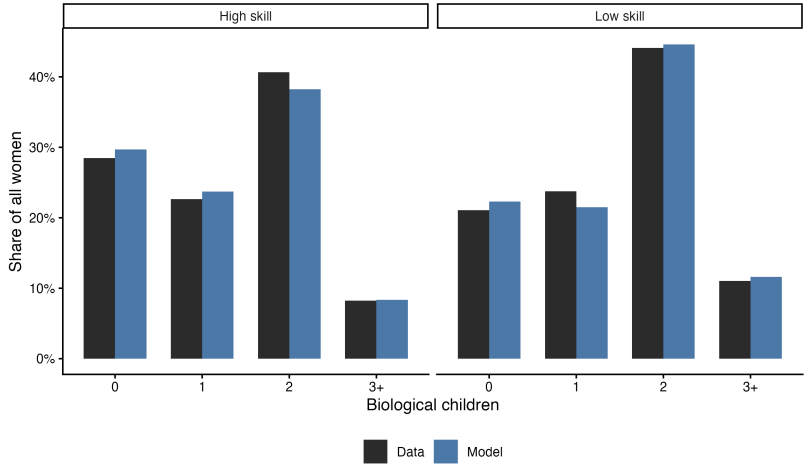
First, compare predicted fertility with the data.

Model fit details →

Outline: Structural Model

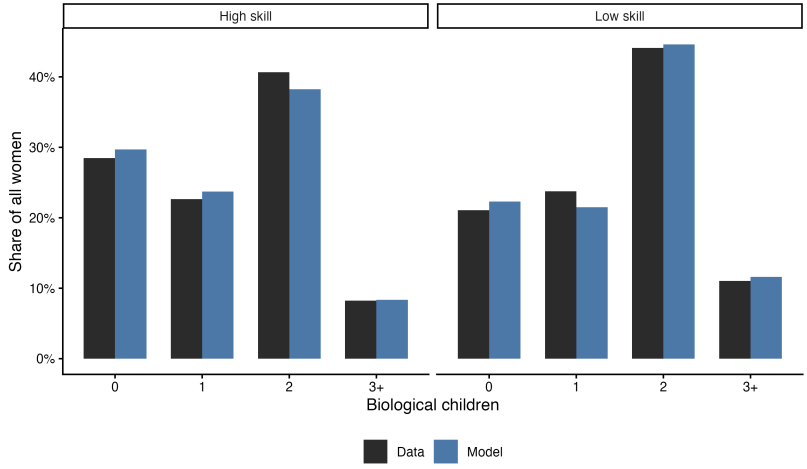
4. Structural Model
- Specification
- Estimation
- Model Fit
- Counterfactuals

Model Fit: Completed Fertility



Full fit and identification →

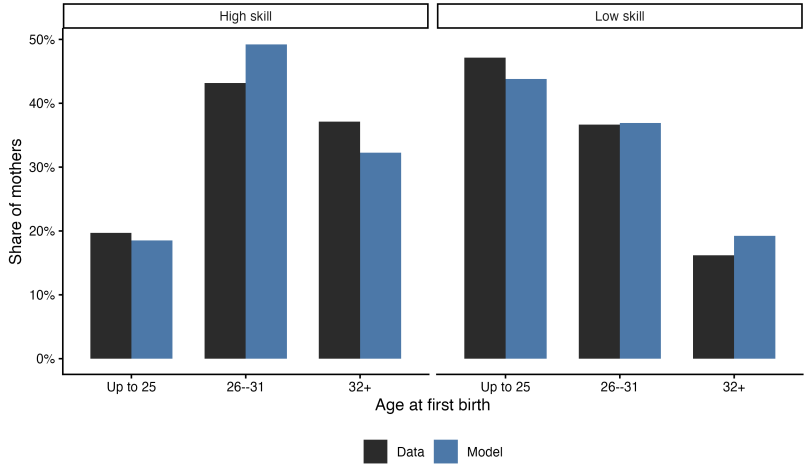
Model Fit: Completed Fertility



Childlessness: model **29.7%** versus **28.5%** in high-skill data; model **22.3%** versus **21.1%** in low-skill data. Shares use all women, including those with no children.

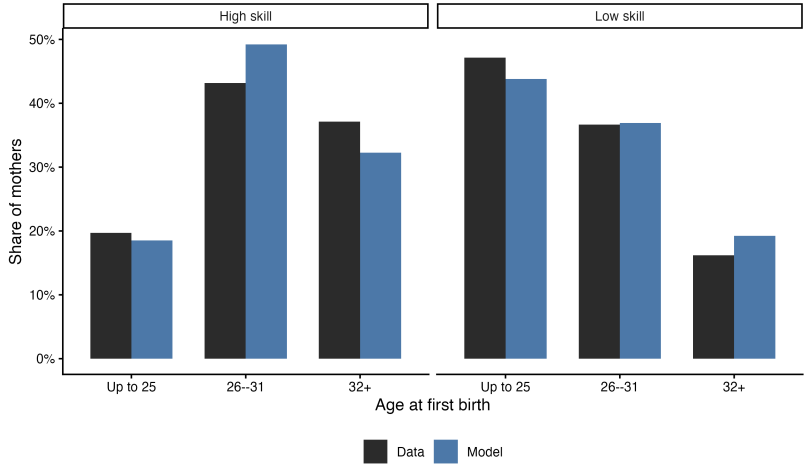
Full fit and identification →

Model Fit: Age at First Birth



Full fit and identification →

Model Fit: Age at First Birth



Average first-birth age: 29.45 for high-skilled women and 26.29 for low-skilled women. Some age groups fit less closely; detailed comparisons are in the appendix.

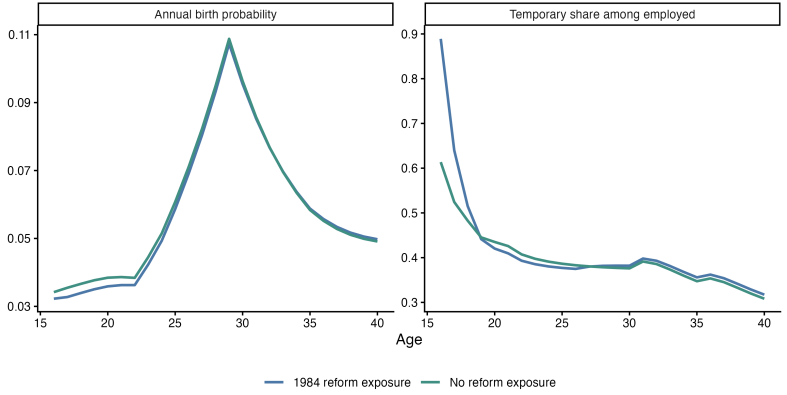
Full fit and identification →

Outline: Structural Model

4. Structural Model
- Specification
- Estimation
- Model Fit
- Counterfactuals

Historical Reform Exposure over the Life Cycle

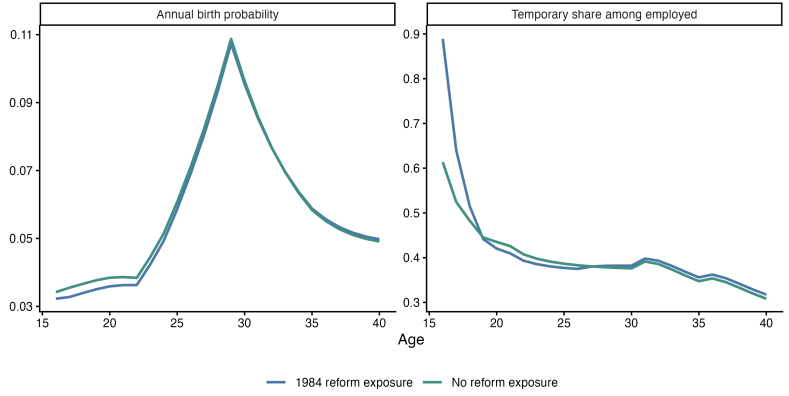
Compare predicted work and fertility with and without reform exposure, holding preferences fixed.



Reduced-form mapping →

Historical Reform Exposure over the Life Cycle

Compare predicted work and fertility with and without reform exposure, holding preferences fixed.



Which channels matter? Next, change one channel at a time and solve women's choices again. The difference shows how much that channel matters in the model.

Reduced-form mapping →

Counterfactual Experiments: Which Channels Matter?

Benchmark: the model's predicted response to the 1984 reform.

Experiment	Δ children per woman	Δ motherhood (pp)	Δ first-birth age (years)	Δ broad index (pp)
Observed 1984 exposure (model fit)	**-0.0280**	**-0.91**	**+0.237**	**+4.15**

Historical benchmark. Compare the 1984 job and resource changes with no reform. The model predicts fewer children and later first births.

Mechanism and sensitivity checks →

Counterfactual Experiments: Which Channels Matter?

Benchmark: the model's predicted response to the 1984 reform.

Experiment	Δ children per woman	Δ motherhood (pp)	Δ first-birth age (years)	Δ broad index (pp)
Observed 1984 exposure (model fit)	**-0.0280**	**-0.91**	**+0.237**	**+4.15**
(a) Starting jobs only	**-0.0298**	**-0.97**	**+0.229**	**+4.28**

(a) Starting jobs only. Change low-skilled women's starting jobs at 16 from outside covered work to temporary work. This produces almost the full fertility decline.

Mechanism and sensitivity checks →

Counterfactual Experiments: Which Channels Matter?

Benchmark: the model's predicted response to the 1984 reform.

Experiment	Δ children per woman	Δ motherhood (pp)	Δ first-birth age (years)	Δ broad index (pp)
Observed 1984 exposure (model fit)	**-0.0280**	**-0.91**	**+0.237**	**+4.15**
(a) Starting jobs only	-0.0298	-0.97	+0.229	+4.28
(b) Later job transitions only	**+0.0026**	**+0.08**	**0.000**	**-0.16**

(b) Later job transitions only. Change only later job prospects. Fertility rises slightly, partly offsetting the starting-job effect. Channel effects can interact.

Mechanism and sensitivity checks →

Counterfactual Experiments: Which Channels Matter?

Benchmark: the model's predicted response to the 1984 reform.

Experiment	Δ children per woman	Δ motherhood (pp)	Δ first-birth age (years)	Δ broad index (pp)
Observed 1984 exposure (model fit)	**-0.0280**	**-0.91**	**+0.237**	**+4.15**
(a) Starting jobs only	-0.0298	-0.97	+0.229	+4.28
(b) Later job transitions only	+0.0026	+0.08	0.000	-0.16
(c) Early resources only	**-0.0003**	**-0.01**	**+0.007**	**0.00**

(c) Early resources only. Reduce resources through age 25 by the estimated reform amount. The small response shows that this loss explains little of the fertility decline.

Mechanism and sensitivity checks →

Counterfactual Experiments: Which Channels Matter?

Benchmark: the model's predicted response to the 1984 reform.

Experiment	Δ children per woman	Δ motherhood (pp)	Δ first-birth age (years)	Δ broad index (pp)
Observed 1984 exposure (model fit)	**-0.0280**	**-0.91**	**+0.237**	**+4.15**
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(b) Later job transitions only	+0.0026	+0.08	0.000	-0.16
(c) Early resources only	-0.0003	-0.01	+0.007	0.00
(d) Equal resources across contracts	**+0.0311**	**+0.95**	**-0.300**	**0.00**

(d) Equal resources across contracts. In the no-reform scenario, give temporary and permanent workers equal resources. Keep job prospects and c_T unchanged. More births and earlier motherhood show that the resource gap matters.

Mechanism and sensitivity checks →

Counterfactual Experiments: Which Channels Matter?

Benchmark: the model's predicted response to the 1984 reform.

Experiment	Δ children per woman	Δ motherhood (pp)	Δ first-birth age (years)	Δ broad index (pp)
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(c) Early resources only	-0.0003	-0.01	+0.007	0.00
(d) Equal resources across contracts	+0.0311	+0.95	-0.300	0.00
(e) No insecurity cost: $c_T = 0$	**-0.0009**	**-0.02**	**+0.004**	**+4.15**

(e) Remove accumulated-insecurity costs. Set $c_T = 0$ in both reform and no-reform scenarios, leaving job paths unchanged. The fertility decline nearly disappears: accumulated insecurity is central in the model.

Mechanism and sensitivity checks →

Counterfactual Experiments: Which Channels Matter?

Benchmark: the model's predicted response to the 1984 reform.

Experiment	Δ children per woman	Δ motherhood (pp)	Δ first-birth age (years)	Δ broad index (pp)
Observed 1984 exposure (model fit)	**-0.0280**	**-0.91**	**+0.237**	**+4.15**
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(c) Early resources only	-0.0003	-0.01	+0.007	0.00
(d) Equal resources across contracts	+0.0311	+0.95	-0.300	0.00
(e) No insecurity cost: $c_T = 0$	-0.0009	-0.02	+0.004	+4.15
(f) Broad insecurity lower by 1 pp	**+0.0085**	**+0.28**	**-0.013**	**-1.00**

(f) Lower broad insecurity by 1 pp. Improve early job prospects to lower broad insecurity by one percentage point. Fertility rises and motherhood starts slightly earlier, holding preferences fixed.

Mechanism and sensitivity checks →

Outline

1. Motivation and Contribution
2. Setting and Design
3. Reduced-Form Results
4. Structural Model
5. **Conclusion**

Conclusion

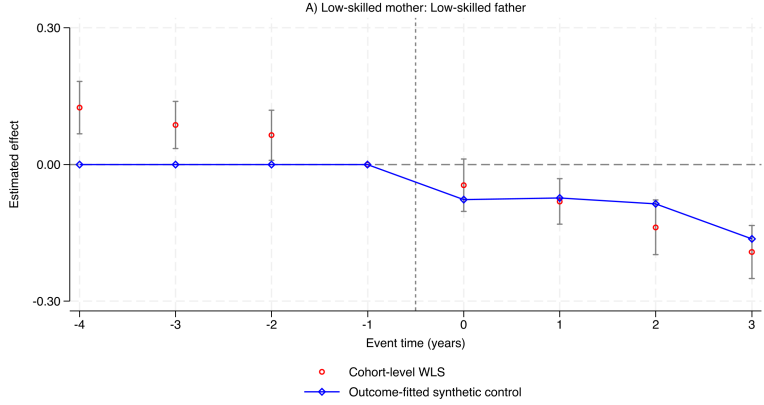
- Reform exposure raises broad **non-permanent employment** and lowers **contribution-based income**, without a detectable change in days worked.
- Birth records show a **15.2%** decline in births per woman; maternal age rises by **0.526 years** and first-delivery age by **0.441 years**.
- Mechanism evidence suggests that employment insecurity constrains their realization, with **exposed women more likely to have fewer children than they desire**.
- **Model:** the fertility decline is driven mainly by **adverse entry conditions** and **accumulated employment insecurity**.

Reforms that increase early-career job insecurity can delay motherhood and reduce realized fertility.

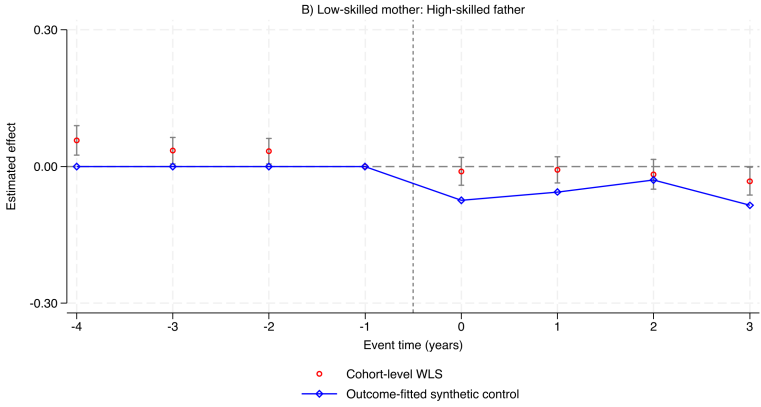
Thank you!

mmartin3@ic.ac.uk

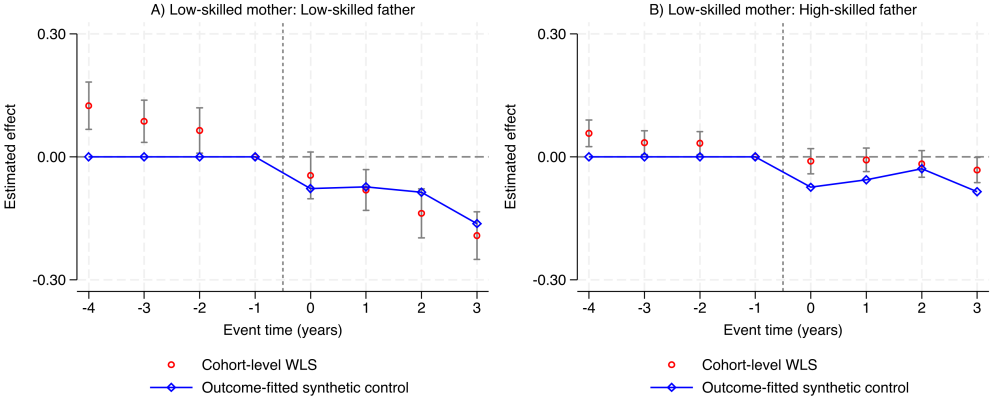
Father-Skill Event Studies



Father-Skill Event Studies



Father-Skill Event Studies



Outline

6. Appendix

Appendix: How the Births-per-Woman Denominator Is Built

$$Y_{cps} = \frac{B_{cps}}{\widehat{D}_{cps}}, \qquad \widehat{D}_{cps} = \underbrace{H_{cp} f_c}_{\text{estimated female birth cohort}} \widehat{\pi}_s.$$

1. **Historical cohort size,** H_{cp}**:** registered births in province p and the mother's birth-cohort quarter c.
2. **Female share,** f_c**:** national female share of births in that cohort year, from historical INE yearbooks.
3. **Skill share,** $\widehat{\pi}_s$**:** respondent-weighted shares in the 1999 Fertility Survey, using 1,337 eligible women born in 1965–1969, **including childless women**.

Low skill: 54.46%; high skill: 45.54%.

The same skill shares are applied to every province and cohort quarter.

B_{cps} counts infants registered in 1980–2017 in the corresponding mother-cohort/province/skill cell. Ceuta and Melilla lack historical denominator support and are excluded from quantity regressions.

Interpretation and weighting →

← Back to talk

Appendix: Interpreting the Cohort Birth Measure

What the denominator achieves

- Scales registered births by an estimated female cohort size, including women who never give birth.
- Keeps the same constructed denominator across the main quantity specifications.

What it does not measure

- Historical female births are not a resident population followed through childbearing; migration and mortality are not adjusted for.
- Survey education proxies recorded occupation. Skill shares are measured in 1999, after the reform, and imposed uniformly across cells.
- The ratio is not completed fertility for linked individual women. Survey observation ages and sampling units differ.

Weighting: descriptive quantity means use $\widehat{D}_{cps}$; DD/event studies use B_{cps}. Percentage effects use the corresponding regression-weighted treated pre mean. Regression standard errors condition on the estimated skill shares.

Survey child-count check →

← Back to talk

Appendix: Fertility Quantity across Datasets

The estimated declines are similar relative to each outcome's pre-reform mean.

	Birth records 1980–2017	Fertility Surveys 1999/2018
Outcome	Births per woman (cohort-level)	Biological children per respondent
Includes childless women	In estimated denominator	Yes, with zero children
DD coefficient	-0.257^{***}	-0.217^{***}
Clustered SE	(0.018)	(0.061)
Treated pre-reform mean	1.693	1.520
Effect relative to pre mean	-15.2%	-14.3%
Observations	3,200 cells	5,621 women

Interpretation: the decline also appears in survey child counts, which do not use the historical cohort denominator.

Notes: Each column retains its own DD sample, weights and geographic effects; surveys include wave effects. Cohort-quarter clustered SEs; *** $p < .01$. This supports the direction of the finding, not equality of estimands. The pooled survey includes women still in their reproductive years, especially in 1999. Childlessness is a separate extensive-margin outcome.

Raw and log birth-count checks →

← Back to talk

Appendix: Perinatal Descriptive Statistics

Birth-record means in the descriptive sample

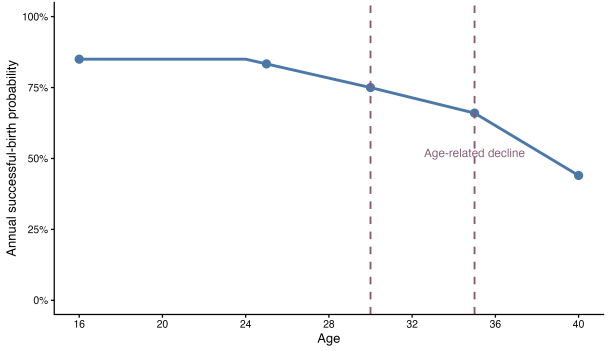
Outcome	Low skill	High skill
Birth weight (grams)	3,224.097	3,234.059
Low birth weight: below 2,500 g (%)	6.8	6.3
High birth weight: above 4,000 g (%)	4.8	4.6
Gestational age (weeks)	38.988	39.067
Preterm birth: below 37 weeks (%)	9.8	7.7
C-section (%)	29.2	36.1

Notes: Unweighted valid infant observations from the manuscript's Table 1, all nationalities, 1980–2017; C-section is available in 2007–2017. These descriptive means cover the birth-record sample; the main perinatal regressions restrict to first deliveries. Multiple-birth infants remain separate records.

← Back to talk

Appendix: Age and Birth Success (Figure G2)

Yes: birth success declines with age in the model.



$$\Pr(\text{birth at age } a \mid x) = q(a)\ \Pr(\text{try at age } a \mid x).$$

$q(a)$ is the annual live-birth probability **conditional on trying**: 0.85 through age 24; 0.75 at 30; 0.66 at 35; 0.44 at 40. Linear interpolation between anchors.

Calibrated before estimation and common across skill and job states. Ages 30/35/40: Leridon (2004); the young-age plateau is a normalization. The model chooses attempts; it does not estimate this schedule from fertility targets.

Annual decision → How much does biological timing explain? →

← Back to talk

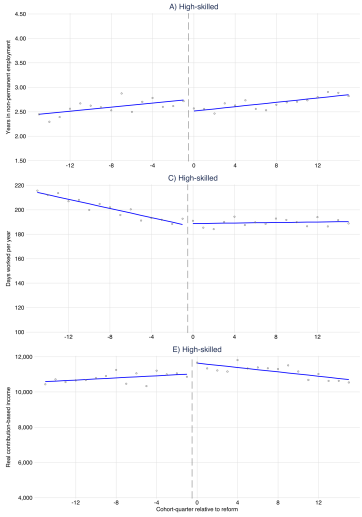
Appendix: Sample and Variable Definitions

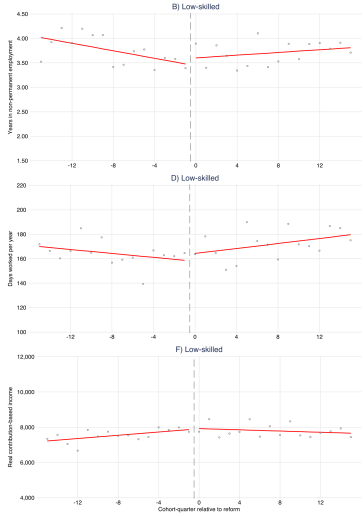
- **Birth records, 1980–2017:** all maternal nationalities. Maternal occupation codes 1–4 define high skill; 5–8 and 10 define low skill; 9, 11, 12 and missing are excluded.
- **Birth quantity:** registered births divided by the maintained historical female cohort stock, allocated with weighted 1999 survey skill shares including childless women. Birth-count regression weights; Ceuta/Melilla lack denominator support.
- **First delivery:** no previous live-born child, using year-specific parity/plurality fields; every infant from a multiple first delivery is retained.
- **Surveys:** official respondent weights. Pooled child counts and 2018 mechanism evidence include childless women; contract and childcare outcomes have specific eligibility restrictions. Model fertility targets use eligible women aged 45–55.
- **MCVL:** source-specific schooling groups; entry-age evidence and career outcomes have distinct affiliation-support restrictions.

Appendix: Career Outcomes and Estimation

- Calendar-age years 16–25 for low-skilled and 18–27 for high-skilled women; at least four years with positive recorded affiliation. Career regressions retain event quarters -15 to 15.
- Non-permanent years use the broad employee-main-affiliation classification, including the specified special employee regimes and the documented missing-contract proxy.
- Employee days count the union of days worked, capped at 365 per year, divided by ten potential years. Overlaps count once; benefit and own-account spells do not count as employee work.
- Income averages complete observed contribution years in 2011 euros. Contribution bases include employment and benefit records; missing years are not assigned zero.
- Regressions allow skill-specific pre/post cohort slopes, include entry-region and birth-quarter effects, and retain regional unemployment assigned to the first retained affiliation year and spell-start quarter. SEs cluster by cohort quarter.

Appendix: First-Decade Career Profiles





Appendix: Birth Counts and Survey Robustness

	Log births per cohort cell	Raw birth count	Biological children per survey respondent
Post × low-skilled	-0.265***	-686.097***	-0.217***
	(0.019)	(60.419)	(0.061)
Mean outcome	7.483	2817.731	1.228
Treated pre-reform mean	7.626	2818.347	1.520
Coefficient (% of pre mean)	—	-24.3	-14.3
Observations	3,328	3,328	5,621
R-squared	0.944	0.950	0.079

← Back to talk

Appendix: Timing Across Regression Estimators

	DD	Event-study
A) Maternal age at birth		
Estimated coefficient	0.526***	0.440***
	(0.067)	(0.088)
Level effect (% of pre mean)	1.9	1.6
Observations	3,382,191	3,382,191
R-squared	0.107	0.108
B) First-delivery age		
Estimated coefficient	0.441***	0.350***
	(0.060)	(0.082)
Level effect (% of pre mean)	1.7	1.4
Observations	1,880,215	1,880,215
R-squared	0.154	0.154

Appendix: Maternal-Age Threshold Estimates

	Pr(age $\leq$20)	Pr(age $\geq$35)	Pr(age $\geq$40)
A) Any birth			
Post $\times$ low-skilled mother	-0.013***	0.015***	0.008***
	(0.001)	(0.004)	(0.002)
Mean outcome	0.080	0.208	0.039
Treated pre-reform mean	0.131	0.134	0.028
Effect (% of pre mean)	-9.9	11.6	29.5
Observations	3,382,191	3,382,191	3,382,191
R-squared	0.048	0.028	0.003
B) First delivery			
Post $\times$ low-skilled mother	-0.016***	0.007**	0.005***
	(0.002)	(0.003)	(0.001)
Mean outcome	0.123	0.129	0.025
Treated pre-reform mean	0.207	0.070	0.015
Effect (% of pre mean)	-7.8	10.0	32.3
Observations	1,880,215	1,880,215	1,880,215
R-squared	0.082	0.026	0.003

Appendix: Survey-Calibrated Selection Sensitivities

- Weighted 2018 survey: all-birth age: 5.204% loss in motherhood; first-delivery age and health: 4.813% loss in a first biological birth during 1980–2017 at ages 15–55. Both use all nationalities and unrestricted child birth countries.
- Conditional selection rates vary by maternal birth quarter, with weakly decreasing individual selection assumed within each quarter. Losses are transported to high-skilled post-reform comparison records within cohort-quarter–province cells. Endpoints account for the signs of DD regression weights.
- Normal sensitivities use comparison selection probabilities of 72.133% for all-birth age and 71.812% for first-delivery outcomes and vary the common correlation over $[-1, 1]$. Binary outcomes use latent thresholds with unrestricted cell correlations.
- These are point sensitivities under transport assumptions across the survey and Birth records, skill groups, and infant records. They are not conventional sharp Lee bounds or confidence intervals.

Appendix: Economic Outcomes and Household Resources

	(a) Married	*(b)* Home ownership	*(c)* Difficulty making ends meet	*(d)* Temporary contract	*(e)* Monthly income ≥1,000 EUR
Exposed cohort × low-skilled	0.018	−0.094	0.065	0.157**	0.103
	(0.101)	(0.070)	(0.096)	(0.071)	(0.096)
Mean dep. var.	0.695	0.798	0.497	0.177	0.387
Treated pre mean	0.744	0.792	0.671	0.247	0.131
Effect (% of pre mean)	2.4	-11.8	9.7	63.7	78.7
R^2	0.026	0.033	0.080	0.041	0.149
Observations	3,611	3,611	3,611	1,812	3,611

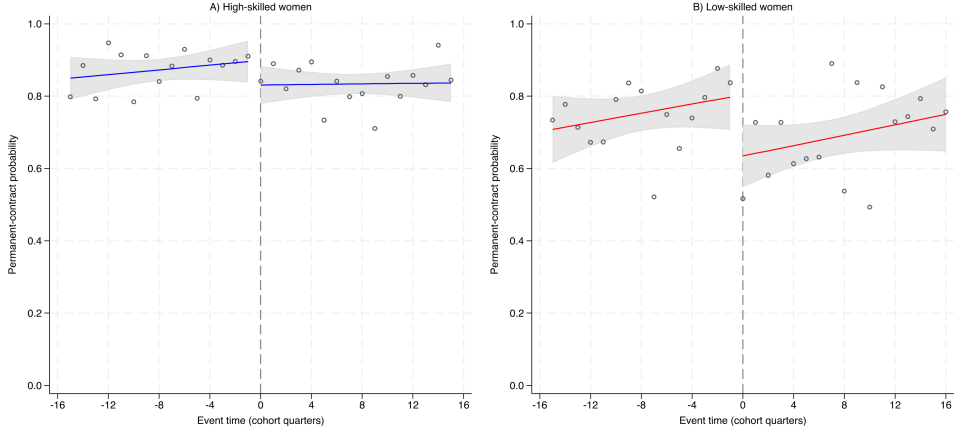
Appendix: Family Structure

	All births		First deliveries	
	Missing father	Single mother	Missing father	Single mother
Exposed cohort × low-skilled	-0.000	0.022***	-0.001**	0.018*
	(0.000)	(0.008)	(0.000)	(0.010)
Treated pre mean	0.021	0.083	0.032	0.127
Mean dep. var.	0.015	0.083	0.023	0.131
Mean dep. var. (treat=1)	0.020	0.079	0.030	0.123
Effect (% of treated pre mean)	-0.1	26.3	-2.9	14.6
Observations (millions)	3.38	0.23	1.88	0.08
R-squared	0.005	0.013	0.007	0.017

Appendix: Preferences, Norms and Support

	(a) Age at economic independence	*(b)* No childcare used (under 14)	*(c)* Unmet desired fertility	*(d)* Egalitarian attitudes	*(e)* No religious affiliation
Exposed cohort × low-skilled	0.760	−0.205	0.279**	0.029	0.019
	(1.224)	(0.231)	(0.106)	(0.031)	(0.070)
Mean dep. var.	22.802	0.278	0.388	0.720	0.174
Treated pre mean	21.874	0.396	0.294	0.691	0.132
Effect (% of pre mean)	3.5	-51.8	95.1	4.3	14.5
R^2	0.047	0.069	0.021	0.057	0.036
Observations	3,057	629	3,611	3,611	3,265

Appendix: Permanent-Contract Profiles



Weighted cohort means and piecewise-linear descriptive fits; women employed with observed contracts. All source-eligible cohorts retained. Source: INE 2018 Fertility Survey, original respondent weights.

Officially weighted 2018 Fertility Survey profiles among employees with observed contracts; descriptive cohort means and linear fits.

Appendix: Specification Robustness

- Post-cutoff cohort-position gradients show larger quantity reductions and longer delays among later cohorts; they do not measure individual temporary-contract exposure.
- Joint pre-period tests: quantity $p = 0.000120$; maternal age $p = 0.204$; first-delivery age $p = 0.025$. These motivate the trend and window checks.
- Quadratic and cubic trends leave the baseline estimates largely unchanged.
- Sixteen-quarter windows yield smaller effects; adding four quarters at each end yields larger effects.
- These specifications retain fewer births and later childbearing, while magnitudes depend on the cohort window.

Fertility Survey comparison → Alternative quantity outcomes →

← Back to talk

Appendix: Structural State, Timing, and Transitions

State.

$$x_a = (s, e_a, n_a, g_a, h_a), \qquad e_a \in \{O, T, P\}, \quad n_a \in \{0, 1, 2, 3\}.$$

- s: skill; e_a: outside covered work, temporary work, or permanent work.
- n_a: parity; g_a: birth spacing; h_a: accumulated temporary-work years, capped at 2.

Transitions.

$$\Pr(e_{a+1} = m \mid e_a = n, s, a) = P_{s,a}(n, m), \qquad n_{a+1} = \min\{n_a + B_a, 3\}.$$

Exposure updates with the next labor state:

$$h_{a+1} = \min\{2, h_a + \mathbf{1}\{e_{a+1} = T\}\}.$$

Each additional temporary-work year adds one, up to two; outside or permanent work leaves h unchanged. MCVL estimates $P_{s,a}$; $q(a)$ is live-birth success after an attempt.

Estimated transition probabilities →

← Back to talk

Appendix: Birth Payoff and Parameter Meanings

$$\begin{aligned}\Delta u_a = \alpha_0 + \alpha_2 \mathbf{1}\{n=1\} + \alpha_3 \mathbf{1}\{n=2\} + \alpha_D D + \alpha_R R + \alpha_G G \\ - c_A(a) - c_H \mathbf{1}\{s=H\}\omega_H(a) - c_T h \\ + \log c_a^1 - \log c_a^0.\end{aligned}$$

- α_0: baseline child value; α_2, α_3: additions for second and third births.
- $c_A(a), c_H$: age costs and the high-skill early-career opportunity cost.
- c_T: utility cost per unit of cumulative temporary exposure h; it persists after the current contract changes.
- $\alpha_D, \alpha_R, \alpha_G$: values attached to centered desired fertility, religiosity, and traditional gender norms.
- D is centered desired family size capped at three; R, G are centered and measured in standard deviations.
- Fixed child cost: $\kappa = 0.06$ (600 euros per child-year). Minimum resources: $\underline{c} = .05$ (500 euros). Resource units: 10,000 euros.

D, R, G are observed preference shifters from the 2018 Fertility Survey, not policy treatments.

← Back to talk

Appendix: Bellman Equation and Exact Solution

$$
\begin{aligned}
v_a^0(x) &= \log c_a^0(x) + \beta E[V_{a+1}(x') \mid x, d = 0], \\
v_a^1(x) &= \log c_a^0(x) + q(a)\Delta u_a(x) + \beta \left[q(a) E\{V_{a+1}(x'_B)\} + (1 - q(a)) E\{V_{a+1}(x'_0)\} \right], \\
V_a(x) &= \log\{\exp v_a^0(x) + \exp v_a^1(x)\}, \\
\Pr(d_a = 1 \mid x) &= \Lambda\{v_a^1(x) - v_a^0(x)\}.
\end{aligned}
$$

- v_a^0, v_a^1: value of not attempting and attempting a birth; V_a: inclusive value before the choice shock.
- β: annual discount factor; $q(a)$: age-specific live-birth success probability; $\Delta u_a(x)$: payoff gain from a successful birth.
- Λ: logistic CDF converting the value difference into an attempt probability; $P_{s,a}$: labor-state transition matrix.
- **Solution:** backward induction from age 40 to 16, then exact forward propagation with low-skill labor entry at 16 and high-skill entry at 18, integrating observed preference types.
- **Output:** model moments are computed without simulated agents or Monte Carlo error.
- Choice shocks are independent type-I extreme value with unit scale. The terminal condition is $V_{41}(x) = 0$.

← Back to talk

Appendix: Estimation Targets and Identification

18 moments discipline 10 preference parameters; labor processes and calibrations are fixed first.

Parameter block	Primary identifying moments
Child/parity values $\alpha_0, \alpha_2, \alpha_3$	Motherhood and unconditional one- and two-child shares by skill
Age/skill costs c_E, c_L, c_H	First-birth age bins, mean first-birth age, and high-low timing differences
Temporary-exposure cost c_T	Labor first-stage mapping plus three reduced-form fertility and timing responses
Preference values $\alpha_D, \alpha_R, \alpha_G$	Conditional 2018 Fertility Survey gradients
Transitions and earnings $P_{s,a}, y_{s,a,e}$	Pre-reform cohort MCVL age-skill-state histories; estimated in the first step
Fecundity and discounting	External $q(a)$, $\beta = 0.96$, and resource normalizations

Parameters, preferences, and diagnostics →

← Back to talk

Appendix: Estimated Structural Parameters

Parameter	Estimate	Standard error
Birth value	-0.511	0.307
Second-birth shift	1.225	0.173
Third-birth shift	-2.800	0.298
Early-age utility shift	1.420	0.234
Late-age cost	1.135	0.141
High-skill career cost	2.466	0.137
Cost per temporary-exposure unit	0.745	0.189
Desired-family-size value	1.710	0.108
Religiosity value	0.088	0.039
Traditional gender-norm value	0.566	0.077

Model questions and evidence →

Appendix: Key Model-Fit Moments

Moment	Data	Model	Standardized residual
Motherhood share, high skill	0.715	0.703	-1.33
Age at first birth, high skill	29.815	29.450	-1.83
Motherhood share, low skill	0.789	0.777	-0.86
Age at first birth, low skill	26.166	26.285	0.59
Desired–realized fertility gradient	0.599	0.619	1.38
Religiosity–fertility gradient	0.032	0.034	0.16
Traditional norms–first-birth-age gradient	-0.658	-0.739	-0.82
1984 effect on capped children	-0.094	-0.028	0.81
1984 effect on motherhood	-0.037	-0.009	0.71
1984 effect on first-birth age	0.441	0.237	-3.41

← Back to talk

Appendix: Numerical Identification Diagnostics

- Eight dispersed starts; all ten preference parameters optimized jointly.
- Objective: **66.45**; 18 targeted moments and 10 estimated parameters.
- Weighted Jacobian rank: **10**; condition number **25.42**.
- Maximum Gauss–Newton correction: 0.0000022.
- Probability mass, scalar/batched solutions, finite differences and preference integration are checked.
- Uncertainty includes shared survey observations, labor first steps, and timing from Birth records.

Numerical convergence does not establish economic fit. The reform timing residual is -3.41 target standard errors.

Model questions and evidence →

← Back to talk

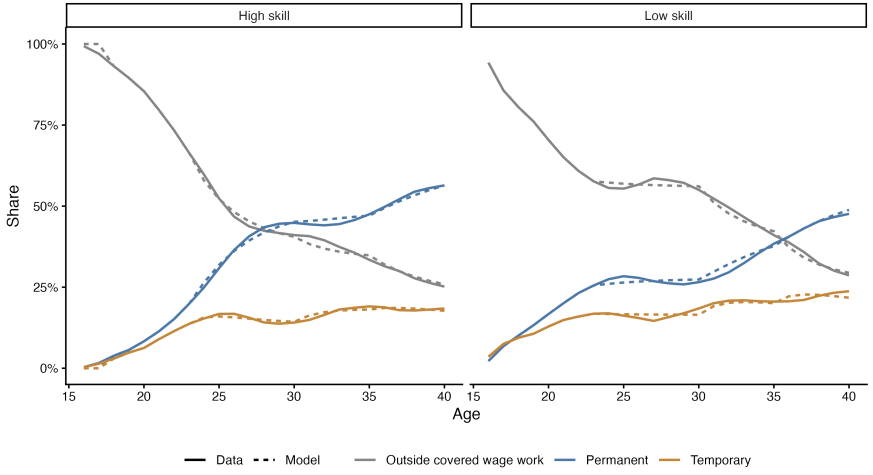
Appendix: Final Empirical Mapping

Input	Code-generated value and model interpretation
Entry	+14.30 pp conditional affiliation-entry DD, transported to temporary entry at age 16 for low skill; high-skill potential entry remains 18.
Early exposure	+0.415 broad-index years in the first decade; own career-sample observation rule.
Permanent access	Signed temporary-retention adjustment 0.0689 (SE 0.0802), disciplined by later stocks and T-to-P flows.
Resources	-535.82 euros of contribution income, mapped proportionally using its regression-sample pre-reform mean.
Fertility targets	Capped children -0.094; motherhood -0.037; first-delivery age +0.441 years.

The affiliation population, selected career sample and income-observed years differ. Conditional selection and cross-source skill transports remain assumptions; the contribution measure is not a wage estimate.

← Back to talk

Appendix: Structural Labor-State Validation



Empirical entry shares at 16 for low skill and 18 for high skill; high-skill women occupy O before 18. Pre-cohort transitions discipline this internal labor-process check.

Model questions and evidence →

Appendix: Estimated Labor-Market First Steps

A. Empirical labor regressions Labor outcome	Estimate	SE	Women	Cohorts
Entry by 19 (percentage points)	14.30	1.53	34,680	44
Broad non-permanent exposure (years/decade)	0.415	0.142	27,191	42
Contribution resources (euros/year)	-535.82	249.86	26,382	42
Employee days (days/year; check)	4.26	6.24	27,026	42
Temporary share, career years 11–18 (pp)	0.37	1.14	50,657	44
T-to-P mobility, career years 11–18 (pp)	-0.52	0.88	29,949	44
Temporary share, 2018 employees (pp)	15.72	7.13	1,812	43
B. Jointly fitted labor-process adjustments **Labor-process parameter**	**Estimate**	**SE**	**MCVL SE**	**Survey SE**
Early T-destination log-odds adjustment	-0.0367	0.0971	0.0970	0.0046
Persistent T/P retention log-odds shift	0.0689	0.0802	0.0795	0.0107
Employed-resource log multiplier	-0.0146	0.0359	0.0359	0.0000

Model questions and evidence →

Appendix: Model Questions and Evidence

Click the question raised by the audience.

Question	Evidence
How do jobs evolve?	Transition probabilities; labor fit
What does the reform change?	First-step estimates; mapping
How are resources measured?	Income profiles and child costs
How do preferences enter?	Survey measures; estimates
What identifies the model?	Targets; numerical checks
Does declining birth success drive the result?	Age profile (G2); decomposition
How sensitive are the results?	Calibrations; persistence and biology
How precise are the counterfactuals?	Standard errors and intervals

Back to estimation →

← Back to talk

Appendix: What Happens after a Temporary Job?

One-year transition probabilities from temporary employment

Skill	Age	To outside	Remain temporary	To permanent
High	16–22	29.7%	51.0%	19.3%
High	23–29	25.8%	50.1%	24.0%
High	30–34	18.7%	66.3%	14.9%
High	35–40	13.4%	72.0%	14.5%
Low	16–22	30.8%	52.4%	16.9%
Low	23–29	31.9%	53.6%	14.5%
Low	30–34	23.2%	62.2%	14.6%
Low	35–40	19.9%	65.6%	14.5%

These probabilities govern the job prospects women anticipate when deciding whether to have a child.

Notes: Adjacent MCVL person-years; baseline age-skill transitions. Outside includes schooling and years without covered wage employment. Rows sum to 100% up to rounding; moving to permanent work can involve a new employer. Fertility does not alter job transitions in the model.

Reform first-step estimates →

Model questions →

Appendix: Resources across Employment States

Median annual contribution-based income (euros)

Skill	Age	Temporary	Permanent	Temp./perm.
High	16–22	2,211	10,804	0.20
High	23–29	7,703	17,227	0.45
High	30–34	10,155	21,789	0.47
High	35–40	14,065	23,391	0.60
Low	16–22	2,810	8,547	0.33
Low	23–29	6,363	11,168	0.57
Low	30–34	6,807	11,724	0.58
Low	35–40	7,802	12,894	0.61

Outside covered work: the larger of 2,000 euros and 40% of the permanent-work profile.

Children: deduct 600 euros per child per year; resources cannot fall below 500 euros.

Notes: Positive-income MCVL person-years. These are contribution-based resource proxies, not hourly wages or causal contract wage gaps. Profiles are estimated before preferences.

Model questions →

Appendix: What Do the Preference Measures Capture?

Measure	Construction	Role in estimation
Desired family size D	Desired children, capped at three and population-centered	Child-number gradient disciplines α_D.
Religiosity R	Standardized affiliation/practice index	Child-number gradient disciplines α_R.
Traditional gender norms G	Standardized seven-item index	First-birth-age gradient disciplines α_G.

$$\text{Preference contribution to birth value} = \alpha_D D + \alpha_R R + \alpha_G G.$$

Interpretation: observed differences in family preferences help explain who has children and when. These fixed survey characteristics are held constant in policy experiments.

Notes: 2018 Fertility Survey measures; coefficients describe conditional gradients, not causal effects of religiosity or gender norms. The structural norms index differs from the reduced-form mechanism index.

Estimated preference parameters →

Model questions →

Appendix: Sensitivity to Calibrated Assumptions

Predicted reform effect relative to the corresponding no-reform baseline

Assumption	Children	Motherhood (pp)	First-birth age
Baseline	-0.028	-0.91	$+0.237$
Double child-resource deduction	-0.022	-0.95	$+0.158$
Discount factor $\beta = 0.94$	-0.030	-0.92	$+0.260$
Discount factor $\beta = 0.98$	-0.026	-0.89	$+0.207$
First stage: lower 95% confidence limit	-0.012	-0.39	$+0.228$
First stage: upper 95% confidence limit	-0.042	-1.36	$+0.243$
Re-estimate all ten preference parameters			
Double child-resource deduction	-0.027	-0.93	$+0.232$
Discount factor $\beta = 0.94$	-0.027	-0.89	$+0.231$
Discount factor $\beta = 0.98$	-0.029	-0.96	$+0.253$

Fewer children and later motherhood remain under these alternative calibrations.

Notes: Preference parameters remain fixed except in the bottom panel. Children are unconditional; first-birth age is conditional on motherhood and measured in years.

Persistence and biological assumptions →

Model questions →

Appendix: Persistence and Biological Assumptions

Assumption	Children	Motherhood (pp)	First-birth age
Baseline	-0.028	-0.91	$+0.237$
Birth success after 35: -20%	-0.028	-0.87	$+0.241$
Birth success after 35: $+20\%$	-0.028	-0.96	$+0.231$
Age-invariant birth-success schedule	-0.030	-1.13	$+0.220$
Biological decline delayed three years	-0.029	-0.99	$+0.234$
Insecurity cost only while currently temporary	-0.002	-0.05	$+0.037$

Mechanism: the result is much smaller when past temporary employment stops affecting birth payoffs after the contract changes. Accumulated insecurity matters within this specification.

Notes: Preference parameters are held fixed. All effects compare reform and no reform under the stated assumption; the reproductive horizon still ends at age 40. First-birth age is in years, conditional on motherhood.

Birth-attempt decomposition →

Model questions →

Appendix: How Precise Are the Counterfactuals?

Sampling uncertainty under the maintained model

Estimand	Estimate	Standard error	95% interval
Historical: completed children	-0.028	0.010	[-0.047, -0.009]
Historical: motherhood	-0.009	0.003	[-0.015, -0.003]
Historical: first-birth age	+0.237	0.048	[+0.143, +0.331]
Local 1 pp: completed children	+0.008	0.002	[+0.005, +0.012]
Local 1 pp: motherhood	+0.003	0.001	[+0.001, +0.004]
Local 1 pp: first-birth age	-0.013	0.002	[-0.017, -0.010]

What enters the intervals? Uncertainty in preference parameters, labor transitions and income profiles, and the estimated reform mapping.

Notes: Motherhood is in probability units here ($0.01 = 1$ pp); first-birth age is in years. "Local 1 pp" fixes the reduction in broad insecurity by construction. Intervals cover sampling uncertainty, not uncertainty across model specifications.

Model questions →

Appendix: Fewer Attempts or Lower Birth Success?

Accounting for the model's reform-induced decline in children

Component	Δ completed children
Change in attempts at peak success rate	-0.027
Age–success-probability interaction	-0.001
Total completed-fertility effect	-0.028

About 95% reflects fewer attempts before age 40. Shifting attempts toward ages with lower birth success accounts for the remainder.

First value the change in total attempts at the peak success probability. Then add the effect of where those attempts fall across ages.

Notes: Unrounded components are -0.0265 and -0.0015 children. This is an accounting decomposition within the model, not a separate causal estimate of biological aging.

Age-specific birth-success profile (G2) →

Model questions →